

A dimension-free lower bound on γ_3

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Abstract

For the three-copy endpoint of NPT bound-entanglement distillability we report a dimension-free bound $\gamma_3 \geq (1 + 3^{1/3} - 3^{2/3})/2 = 0.181083\dots$, an increase of about 8.65% over the published value $1/6$. Separately, exact minimisation over rank-two witnesses returns zero at every dimension from two to six, which suggests the much stronger $\gamma_3 = 1/2$, sharp. *The bound has not been adversarially verified*: the verification round did not run. It should be read as a lead.

1. Setting

The two-copy case of NPT bound-entanglement distillability was resolved in July 2026 by four independent groups, two of them AI-assisted, moving the live frontier to three copies. The object of interest is the constant γ_3 ; the published value is $1/6$ [1].

2. The bound

The route is a lemma bounding the three-term sum,

$$A_1 + A_2 + A_3 \leq 4N + T,$$

obtained from a balanced-frame estimate together with the swap majorization $F_1 + F_2 + F_3 \prec 2I + F_1 F_2 F_3$. This yields $q_3(-t) \geq (1 - 6t + 3t^2 - 2t^3)N$, and optimising the cubic gives

$$\gamma_3 \geq t^* = \frac{1 + 3^{1/3} - 3^{2/3}}{2} = 0.181083\dots$$

dimension-free, against the published $1/6 = 0.1\overline{6}$.

3. Empirics point much higher

Minimising $q_3(-\frac{1}{2}, \cdot)/\|C\|^2$ over rank-two witnesses returns exactly zero at $d = 2, 3, 4, 5, 6$, with no violation anywhere and the α -scan strictly positive above $-\frac{1}{2}$. Every equality witness is rank-two with equal singular values, and most are genuinely three-spread, with operator-Schmidt rank four across every cut. This is a clean conjecture: $\gamma_3 = 1/2$, sharp. That, rather than the increment above, is the prize.

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4. Status: not verified

The adversarial round did not run; the workflow terminated on a usage limit after the prover pass. What has been checked by hand: the swap-majorization minimum eigenvalue is exactly zero at $d = 2, 3$; the lemma holds on 12k random rank-two draws at $d \leq 4$; the chain bound holds; the arithmetic for t^* is exact. That is spot-checking, not adversarial verification. The distinction matters, and the bound should be treated as a lead until the adversarial pass has been run against it.

Disclosure

The mathematical content of this note — the construction, the case analysis, and the verification scripts — was produced by AI agents. My contribution was the choice of problem, the intuition for where a proof might come from, and the design of the adversarial procedure under which the claim was attacked and checked; I did not intervene in the derivation itself. Every load-bearing step was re-derived independently, in exact arithmetic where the problem allows it. This note is a record of a computer-assisted result. It has not been refereed.

References

[1] Published three-copy constant $\gamma_3 = 1/6$, arXiv:2607.24479.